

hours (TWh) of incremental power by 2030^{iv}. Business leaders and regulators alike are increasingly concerned with the carbon emissions and energy efficiency of IT operations.

- India's data centers are embracing **liquid cooling, renewables, and cloud migration** to boost energy efficiency.
- The rise of **Green Cloud** practices—like auto-scaling and idle workload shutdowns—is cutting both carbon and costs.
- CIOs now track **CO₂ per transaction** and energy per AI model, aligning IT with ESG goals and rising energy risks.
- At Cognizant, we have reduced the emissions associated with IT infrastructure by 60% since 2019 by **shifting our focus** from on-site data centers to the cloud, encouraging the use of virtual collaboration tools and eliminating waste in end-user computing behaviors^v.

In short, *building resilience and sustainability* into infrastructure is now seen as key to “future-proofing” it. A sustainable infrastructure is not only good for the planet but tends to be more robust (e.g., less likely to overheat or suffer power outages) and cost-effective in the long run. Going forward, winning enterprises will likely be those that manage to scale their digital capabilities **while shrinking the energy and carbon impact** through clever engineering and continuous optimization.

The Road Ahead – Adapting for an AI-First Era:

For CIOs and IT leaders, the challenge now is **operationalizing** these trends. How can organizations pivot from traditional setups to AI-first, cloud-smart, sustainable infrastructure? Cognizant recommends a phased transformation roadmap:

- **Immediate (0–3 months): Assess and strategize.** Take stock of current infrastructure and its readiness for AI. Audit where your data lives and evaluate compliance requirements. Identify quick wins – e.g., consolidating unused servers

or enabling basic cloud services – to start building momentum. Set clear goals, such as “enable GPU cloud for data science team within 3 months” or “cut legacy hosting costs by 20%.”

- **Near-term (3–6 months): Pilot and secure.** Experiment with multi-cloud by moving a non-critical workload to a second cloud provider or deploying a new app on a domestic (India-local) cloud to test latency and integration. Implement **Zero Trust security** frameworks to tighten access control across your now-spread-out infrastructure. Also set up **observability tools** – integrated dashboards and AIOps platforms – to get visibility into how systems are performing across on-prem and cloud. This is also the phase to introduce **AI in IT operations** on a trial basis: for example, use an AI-based monitoring tool that can predict and auto-remediate common server issues.

- **Mid-term (6–12+ months): Scale and institutionalize.** With lessons from pilots, start rolling out a unified *infrastructure platform*. This means providing your developers and business units a self-service catalog of infrastructure services (cloud environments, containers, database instances, etc.) with defined **SLAs and cost controls**. Establish a FinOps practice to continuously track usage and spend. Importantly, invest in **talent upskilling and process change**: train your IT staff in cloud architecture, AI/ML basics, and new tools. Break down silos between infrastructure, development, and analytics teams – perhaps create a cross-functional “AI Infrastructure” center of excellence. Cognizant often aids clients at this stage by running **talent transformation programs** – for instance, it recently conducted “**Vibe Coding Week**,” a company-wide AI hackathon event with over 250,000 employees, to accelerate AI literacy across the organization. Such initiatives can be emulated to build an AI-ready culture in IT teams. By the end of the year, aim to have

infrastructure operations functioning like a product organization: regularly introducing improvements, closely aligning capacity with business demand, and orchestrating resources across cloud and on-premise seamlessly.

Throughout these steps, Cognizant works as a partner to its clients, bringing templates, playbooks and experts for each stage – whether it's assessing an SAP landscape for cloud migration, or designing a sovereign cloud solution for a bank, or implementing an AIOps toolchain. This holistic approach ensures companies aren't left to figure it out alone; they leverage Cognizant's accumulated best practices and solution accelerators.

Conclusion:

The landscape of enterprise IT infrastructure in India is being reshaped at a breakneck pace by the twin engines of AI and cloud. Organizations that successfully ride this wave stand to gain massively in terms of agility, innovation, and market leadership. India is uniquely positioned with its strong tech talent and late-mover advantage to jump ahead in this journey.

Cognizant's vision is clear: the future is **AI-first, powered by cloud-smart infrastructure**, executed with an agile product mindset and responsible practices. In this future, **GenAI is the spark, infrastructure is the engine, adaptive governance is the steering wheel, and talent is the fuel** driving India Inc. forward.

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